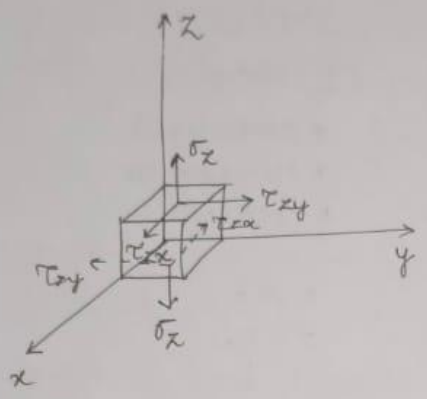
Brittle v/s ductility

→ factors :

* Temperature (brittle → roomtemp. Ductile → high temp)
* Loading condition (brittle → Tension ductile → compression)
* Deformation or strain rate



$\tau_{xy} = \tau_{yx}$

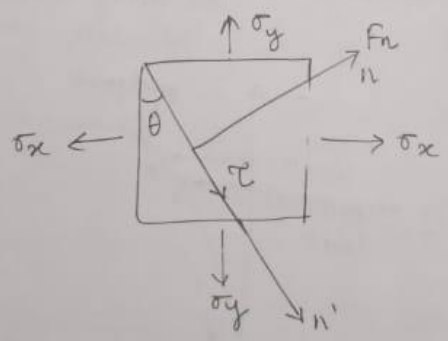
$\tau_{yz} = \tau_{zy}$

$\tau_{xz} = \tau_{zx}$

plane direction

$$\sigma_n = \frac{\sigma_x + \sigma_y}{2} + \frac{\sigma_x - \sigma_y}{2} \cos 2\theta - \tau_{xy} \sin 2\theta$$

$$\tau = \frac{\sigma_x - \sigma_y}{2} \sin 2\theta + \tau_{xy} \sin 2\theta$$



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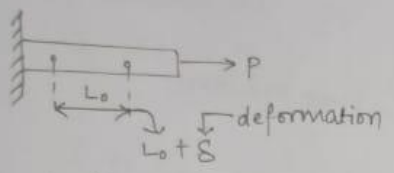
Strength of materials

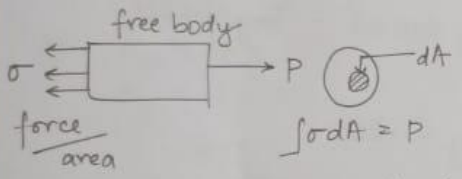
→ External loading
→ Internal resistance
→ Deformation

Hooke's law:

stress is proportional to strain



$$\text{strain} = \frac{\delta}{L_0}$$



$$\int \sigma \, dA = P$$

Assume σ = constant

$$P = \sigma A$$

↓

stress

Brittle : fails immediately after yield point

Ductile : first has plastic deformation

Assumption :

(I) static

static equilibrium :

$$\Sigma F_x = \Sigma F_y = \Sigma F_z = 0$$

$$\Sigma M = 0$$

(II) Body :

* Continuous
* Homogenous
* Isotropic

$$\sigma \propto \varepsilon$$

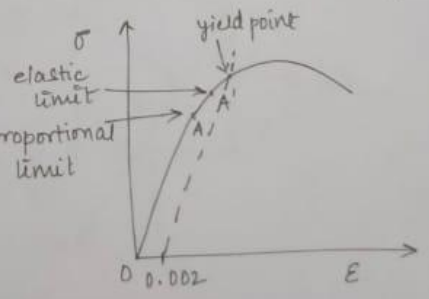
$$\sigma = E\varepsilon$$

↓

Young's modulus

Tensile Test

We apply load and calculate elongation



For $\sigma_{max/min}$:

$$\frac{d\sigma_n}{d\theta} = 0$$

$$\frac{d\sigma_n}{d\theta} = -\sin 2\theta\,(\sigma_x - \sigma_y) - 2\tau_{xy}\cos 2\theta = 0$$

$$\boxed{\tan 2\theta = \frac{-2\tau_{xy}}{\sigma_x - \sigma_y}}$$

→ SAME

for $\tau = 0$: $\boxed{\tan 2\theta = \frac{-2\tau_{xy}}{\sigma_x - \sigma_y}}$

When σ_{max} or σ_{min} → $\tau = 0$: Principal plane

↓

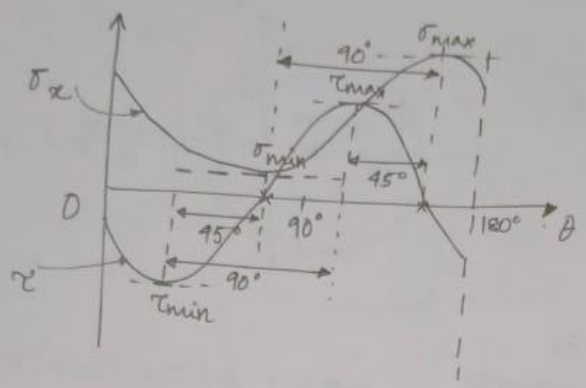
Principal stresses (occur at 90° to each other)

For $\tau_{max/min}$: $\frac{d\tau}{d\theta} = 0$

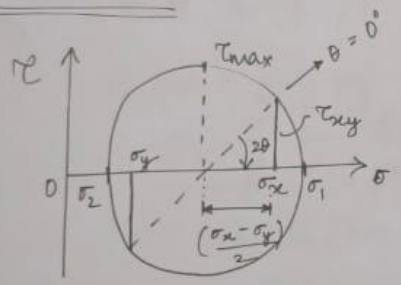
$$\frac{(\sigma_x - \sigma_y)}{\cancel{2}}(\cancel{2}\cos 2\theta) - 2\tau_{xy}\sin 2\theta = 0$$

$$\boxed{\tan 2\theta = \frac{\sigma_x - \sigma_y}{-2\tau_{xy}}}$$

* We can see that $\tau_{max/min}$ occur at $45°$ to the $\sigma_{max/min}$.
* Also τ_{max} & τ_{min} are separated by $90°$
* —"— σ_{max} & σ_{min} ——"——



Mohr's Circle:



$\sigma_{max} = \sigma_1$

$\sigma_{min} = \sigma_2$

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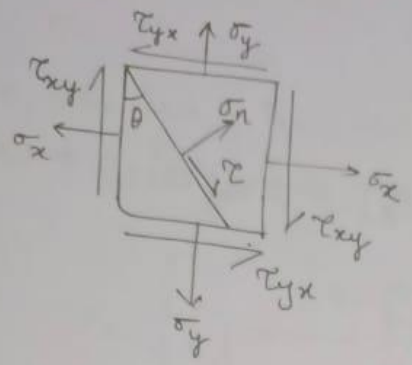

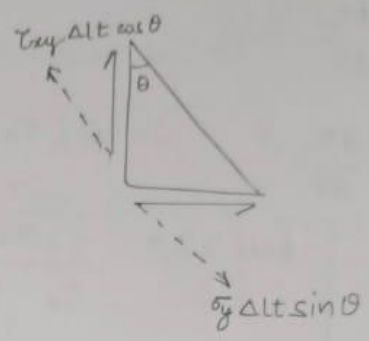

CW rotation : +ve

$$\sigma_n = \left(\frac{\sigma_x+\sigma_y}{2}\right) + \left(\frac{\sigma_x-\sigma_y}{2}\right)\cos 2\theta - \tau_{xy}\sin 2\theta$$

$$\tau = \left(\frac{\sigma_x-\sigma_y}{2}\right)\sin 2\theta + \tau_{xy}\cos 2\theta$$

Q. $\sigma_x = 25p$ MPa $\quad \sigma_n = 50$ MPa

$\sigma_y = 5p$ MPa $\quad \tau = 5$ MPa

$\theta = 45^\circ$

$$50 = \frac{30p}{2} + \cancel{\frac{20p}{2}\cos 90^\circ}^{\,0} - \tau_{xy}\,\cancel{\sin 90^\circ}^{\,1}$$

$$50 = 15p - \tau_{xy} \longrightarrow (1)$$

$$5 = \frac{20p}{2}\sin 90^\circ + \cancel{\tau_{xy}\cos 90^\circ}^{\,0}$$

$$p = 0.5 \rightarrow (2)$$

$$\tau_{xy} = -42.5 \text{ MPa}$$

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$$\begin{vmatrix} \sigma_x - \sigma & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_y - \sigma & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_z - \sigma \end{vmatrix} = 0$$

$$\sigma^3 - \overbrace{(\sigma_x + \sigma_y + \sigma_z)}^{I_1}\sigma^2 + \overbrace{\left\{ \begin{vmatrix} \sigma_x & \tau_{xy} \\ \tau_{xy} & \sigma_y \end{vmatrix} + \begin{vmatrix} \sigma_y & \tau_{yz} \\ \tau_{yz} & \sigma_z \end{vmatrix} + \begin{vmatrix} \sigma_x & \tau_{xz} \\ \tau_{xz} & \sigma_z \end{vmatrix} \right\}}^{I_2} \sigma - \underbrace{\begin{vmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_y & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_z \end{vmatrix}}_{I_3} = 0$$

$$\sigma^3 - I_1\sigma^2 + I_2\sigma - I_3 = 0$$

$\begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \end{Bmatrix} \Rightarrow$ Sol^{n}s are principal stresses

→ They are independent of choice of reference frame

→ Depends on state of stress ~~of~~ at a point.

$$\sigma_{ij} = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix}$$

$I_1 = \sigma_1 + \sigma_2 + \sigma_3 = \sigma_x + \sigma_y + \sigma_z$

$I_2 = \sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1$

$I_3 = \sigma_1\sigma_2\sigma_3$

Q. $\sigma_{ij} = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ $\quad \sigma_{1-3} = ?$

$I_1 = 3$ $\qquad \sigma^3 - 3\sigma^2 - 3\sigma + 1 = 0$

$I_2 = -3$ $\qquad (\sigma + 1)(\sigma^2 - 4\sigma + 1) = 0$

$I_3 = -1$ $\qquad \sigma_1 = -1$ and

$\sigma_2 = 2 + \sqrt{3}$ and $\sigma_3 = 2 - \sqrt{3}$

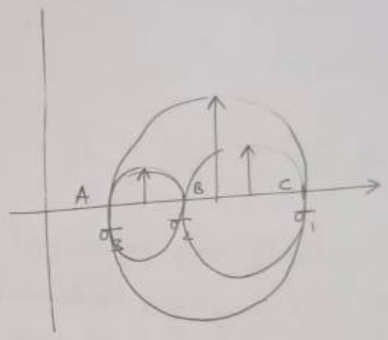
Mohr's circle for 3D:

(1) A, B, C are principal stresses

(2) 3 extreme τ,

$\frac{\sigma_1 - \sigma_2}{2}$, $\frac{\sigma_2 - \sigma_3}{2}$

and $\frac{\sigma_3 - \sigma_1}{2}$



(3) $\tau_{max} = \frac{\sigma_1 - \sigma_3}{2}$ for $\sigma = \frac{\sigma_1 + \sigma_3}{2}$

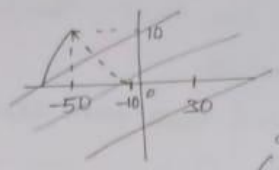
(4) $\sigma_1 = \sigma_2 \neq \sigma_3$ or $\sigma_1 \neq \sigma_2 = \sigma_3$

(5) $\sigma_1 = \sigma_2 = \sigma_3$ (hydrostatic)

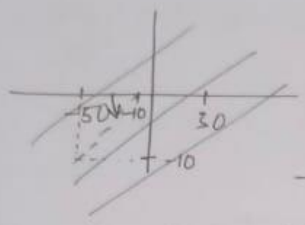
Q. $\sigma_x = -50\,MPa$

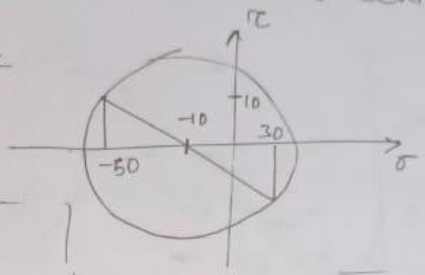
$\sigma_y = 30\,MPa$

$\tau_{xy} = 10\,MPa$



in this CCW: +ve





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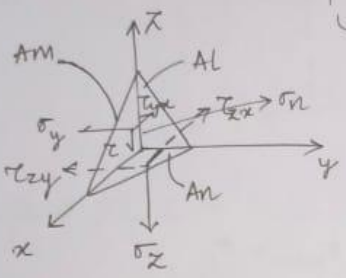


$A\sigma_n = A(\sigma L + \sigma m + \sigma n)$

L, m, n 走 → direction cosines

$$\sigma m - \tau_{zy} n - \sigma_y m - \tau_{xy} L = 0$$

$$\sigma L - \tau_{yx} m - \tau_{zx} n - \sigma_x L = 0$$

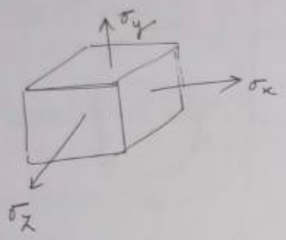
$$\sigma n - \tau_{yz} m - \tau_{xz} L - \sigma_z n = 0$$

Pure Shear

$$\sigma_{ij} = \begin{bmatrix} 0 & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & 0 & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & 0 \end{bmatrix}$$

$$I_1 = 0$$

Hooke's law in 3D:



$$\sigma_x = E\varepsilon_x$$

$$\varepsilon_y = \varepsilon_z = -\nu\varepsilon_x$$

where, $\nu > 0$

Stress	Strain: x-dir^n	Strain: y-dir^n	Strain: z-dir^n
σ_x	σ_x/E	$-\nu\sigma_x/E$	$\frac{-\nu\sigma_x}{E}$
σ_y	$\frac{-\nu\sigma_y}{E}$	$\frac{\sigma_y}{E}$	$\frac{-\nu\sigma_y}{E}$
σ_z	$\frac{-\nu\sigma_z}{E}$	$\frac{-\nu\sigma_z}{E}$	$\frac{\sigma_z}{E}$

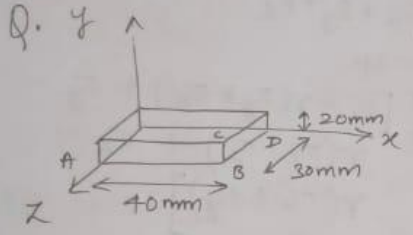
Generalised Hooke's law for homogenous & isotropic material:

$$\varepsilon_x = \frac{1}{E}\left(\sigma_x - \nu(\sigma_y + \sigma_z)\right)$$

$$\varepsilon_y = \frac{1}{E}\left(\sigma_y - \nu(\sigma_x + \sigma_z)\right)$$

$$\varepsilon_z = \frac{1}{E}\left(\sigma_z - \nu(\sigma_x + \sigma_y)\right)$$

Q.



$\nu = 0.3$

$E = 200\,GPa$

$\Delta AB = -0.012\,mm = \varepsilon_x$

Uniform pressure is applied.

$\therefore\ \sigma_x = \sigma_y = \sigma_z = -p$

$$\frac{-0.012\,mm}{40\,mm} = \frac{1}{200\times10^9\,Pa}\left(-p - 0.3(-2p)\right)$$

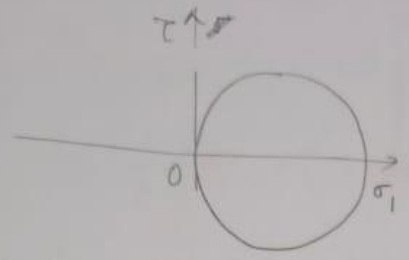
$$-6\times10^7\,Pa = -p + 0.6p = -0.4p$$

$$p = +150\,MPa$$

$\varepsilon_z = -0.009\,mm$ $\quad$ $\varepsilon_y = -0.006\,mm$

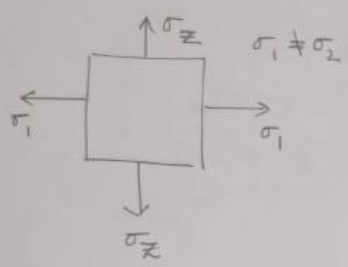
Uniaxial Tension

$\sigma_2 = \sigma_3 = 0$

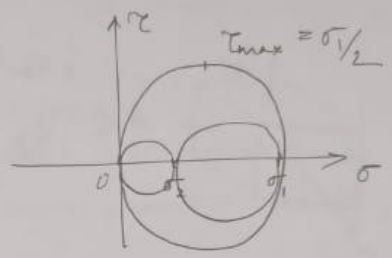


If it had been compression, $\sigma_1 = -ve$.

Biaxial Tension



$\sigma_1 \neq \sigma_2$



Q. $\begin{bmatrix} 0 & -240 & 0 \\ -240 & 200 & 0 \\ 0 & 0 & -280 \end{bmatrix}$ MPa : σ_{ij}

$I_1 = -80$

$I_2 = \begin{vmatrix} 0 & -240 \\ -240 & \cancel{0}\ 200 \end{vmatrix} + \begin{vmatrix} 200 & 0 \\ 0 & \cancel{200}\ -280 \end{vmatrix} + \begin{vmatrix} \cancel{-280}\ 0 & 0 \\ 0 & -280 \end{vmatrix}$

$= \cancel{60800}\ -113600$

$I_3 = -56000$

$\sigma^3 - 60800\sigma^2 - 56000\sigma + 56000 = 0$

$\sigma_1 = 60800.92$ MPa $\quad \sigma_2 = 0.604$ MPa $\quad \sigma_3 = -1.52$ MPa

Q. Continue w/ last Q. given $p = 180 MPa$, find ΔV.

$$k = \frac{200 GPa}{3(1-2(0.3))} = 166$$

$$k = -\frac{p}{e_v} \qquad e_v = \frac{-180 MPa}{166} = \frac{\Delta V}{V_0}$$

$$\frac{\Delta V}{V_0} = -1.08 \times 10^{6}$$

$$\Delta V = -26 mm^3$$

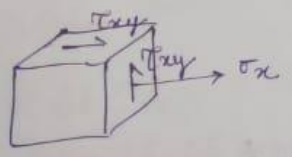
* For shear loading, block changes shape but volume remains constant.

Shear stress:

shear stress

$$\tau = \gamma G \rightarrow \text{shear modulus}$$

shear strain

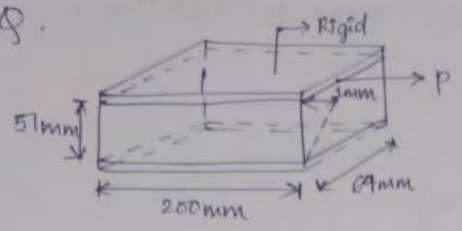


$$\left.\begin{aligned} \tau_{xy} &= \gamma_{xy} G \\ \tau_{yz} &= \gamma_{yz} G \\ \tau_{zx} &= \gamma_{zx} G \end{aligned}\right\} \text{Eq}^{n}\text{s. } \underline{\underline{4-6}}$$

Generalised Hooke's law for general loading condition.

$$\left\{\begin{aligned} 1-3 &\rightarrow \varepsilon_x = \\ &\quad\ \varepsilon_y = \\ &\quad\ \varepsilon_z = \\ 4-6 &\rightarrow \tau_{xy} = \\ &\quad\ \tau_{yz} = \\ &\quad\ \tau_{xz} = \end{aligned}\right.$$

Q.



$G = 0.62\,GPa$

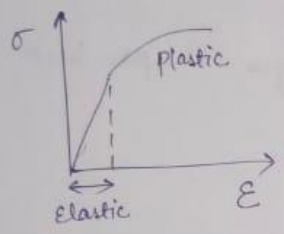
Find: (i) average shear strain

(ii) P

$$\gamma_{xy} = \frac{1mm}{51mm} \simeq 0.02\,rad$$

$$\tau_{xy} = 0.012\,GPa = 12\,MPa$$

$$P = \tau_{xy} \times 200mm \times 64mm = 155.61\,kN$$



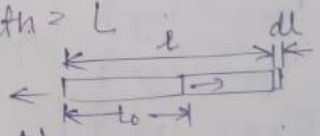
In plastic zone:

(1) Hooke's law ain't valid
(2) Not reversible
(3) Path dependent.
(4) No constant such as young's modulus to connect stress & strain.

$$\varepsilon = \ln \frac{L}{L_0}$$

Initial length of zero stress = L_0

In plastic ~~zone~~ zone, length = L



True strain = ε

Engineering strain = $e = \frac{\Delta L}{L_0}$

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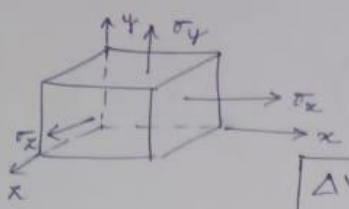


$$V' = (x + x\varepsilon_x)(y + y\varepsilon_y)(z + z\varepsilon_z)$$

$$V' \simeq xyz(1 + \varepsilon_x + \varepsilon_y + \varepsilon_z)$$

$$\boxed{\Delta V = xyz(\varepsilon_x + \varepsilon_y + \varepsilon_z)}$$

$$e_v = \frac{\Delta V}{V_0} = \varepsilon_x + \varepsilon_y + \varepsilon_z$$

for, Hydrostatic loading, $\sigma_x = \sigma_y = \sigma_z = -p$

as $e_v = \varepsilon_x + \varepsilon_y + \varepsilon_z$

$\cancel{e_v = 3p}$

$$e_v = \frac{1}{E}\left[\sigma_x - \nu(\sigma_y + \sigma_z) + \sigma_y - \nu(\sigma_x + \sigma_z) + \sigma_z - \nu(\sigma_x + \sigma_y)\right]$$

$$e_v = \frac{1 - 2\nu}{E}(\sigma_x + \sigma_y + \sigma_z)$$

$$e_v = -\left(\frac{1 - 2\nu}{E}\right)(3p)$$

e_v is always negative and hence k is always +ve.

$\therefore\ 1 - 2\nu > 0$

$\therefore\ 0 < \nu < 1/2$

$$\boxed{\frac{-p}{e_v} = \frac{E}{3(1-2\nu)} = k}$$

where k = $\underbrace{\text{Bulk's modulus}}_{\text{Resistance to volumetric change under Hydrostatic loading.}}$

(iii) Total true strain $= \ln \frac{L_o'''}{L_o} = \underline{0.285}$

Volume change in plastic deformation:

$$\Delta V = xyz(1+e_x)(1+e_y)(1+e_z) - xyz$$

$\downarrow$ engineering strain

$$e_v = \frac{\Delta V}{V_o} = (1+e_x)(1+e_y)(1+e_z) - 1$$

$$\ln e_v = \ln\left((1+e_x)(1+e_y)(1+e_z) - 1\right)$$

$$= \cancel{\ln(1 + e_x + e_y + e_z + e_x e_y + e_x e_z + e_y e_z + e_x e_y e_z - 1)}$$

$$\ln(1+e_v) = \ln(1+e_x) + \ln(1+e_y) + \ln(1+e_z)$$

$$\ln(1+e_v) = \varepsilon_x + \varepsilon_y + \varepsilon_z$$

During plastic deformation, $\Delta V = 0$

$$\therefore \boxed{\varepsilon_x + \varepsilon_y + \varepsilon_z = 0}$$

$$\boxed{\text{also } \nu = 0.5 \text{ for plastic zone}}$$

$$\boxed{e = \frac{\Delta l}{l_0}} \quad \varepsilon = \ln\left(\frac{l_0 + \Delta L}{l_0}\right) = \ln(1+e)$$

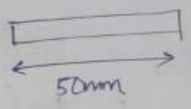
$$\underline{\varepsilon = \ln(1+e)}$$

Q.

True strain	0.01	0.1	0.2	0.5	1	4
Eng. strain	0.01005	0.1052	0.22	0.649	1.72	53.

Elastic zone — Plastic zone

Q. Deformed in 3 increments each case Eng. strain = 0.1



Find (i) Total Eng. strain.

$L_0 = 50mm \quad e = 0.1$, $\Delta L = 5mm$ $\Big\}$ $L_0' = 55mm$

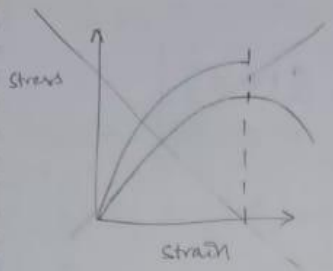
$L_0 = 55mm \quad e = 0.1$, $\Delta L = \cancel{60.5}\ 5.5\ mm$ $\Big\}$ $L_0'' = 60.5mm$

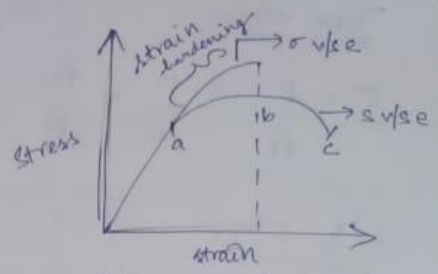
$L_0 = 60.5\ mm \quad e = 0.1$, $\Delta L = 6.05\ mm$ $\Big\}$ $L_0''' = 66.55\ mm$

Total engg. strain = 0.331

(ii) Incremental true strain

$\varepsilon_1 = \ln \frac{L_0'}{L_0} \quad \varepsilon_2 = \ln \frac{L_0''}{L_0'} \quad \varepsilon_3 = \ln \frac{L_0'''}{L_0''}$



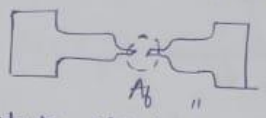


* Conversion formulas are valid only upto 'b'

after point 'b' "necking" starts.

Necking:

elongation occurs only is specific part



* At 'c' fracture occurs,

$$\varepsilon_f = \ln \frac{A_0}{A_f} \qquad \sigma_f = \frac{F_f}{A_f}$$

$$\varepsilon = \ln \frac{l}{l_0}$$

" A_f is area where fracture occurs "

* Hollomon power law (describes σ v/s e curve upto 'b')

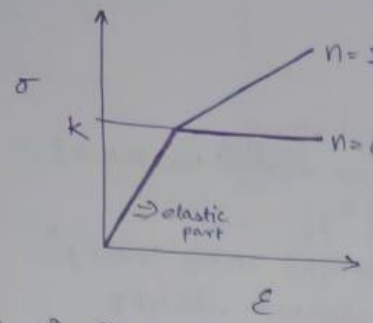
$$\sigma = K\varepsilon^n$$

↑ true stress, ↑ true strain

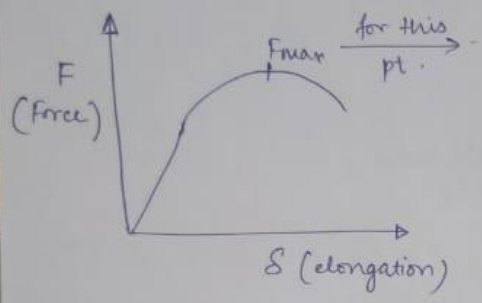
K = strength coefficient

n = strain hardening exponent

$n \in (0, 1)$



* Plastic instability



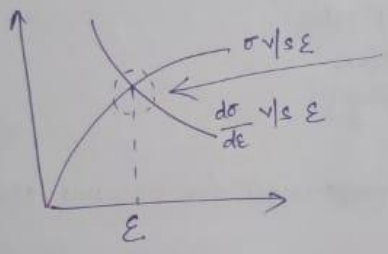
$$F = \sigma A$$

$$dF = 0$$

$$\sigma dA + A d\sigma = 0$$

$$\frac{d\sigma}{\sigma} = -\frac{dA}{A} = d\varepsilon$$

$$\boxed{\frac{d\sigma}{d\varepsilon} = \sigma}$$



"Considere criteria"

using hollomon's power law,

$$\sigma = K\varepsilon^n$$

$$\frac{d\sigma}{d\varepsilon} = nK\varepsilon^{n-1} = \frac{nK\varepsilon^n}{\varepsilon} = \frac{n\sigma}{\varepsilon}$$

at point of instability, $\frac{d\sigma}{d\varepsilon} = \sigma = \frac{n\sigma}{\varepsilon} \Rightarrow \boxed{n = \varepsilon}$

True stress

$$\sigma = \frac{P}{A} = \frac{P}{A_0} \times \frac{A_0}{A} = \boxed{S \cdot (1+e)}$$

Eng. stress

$$\boxed{S = \frac{P}{A_0}}$$

$A_0 L_0 = A L$

$$\boxed{\varepsilon = \ln(1+e)}$$

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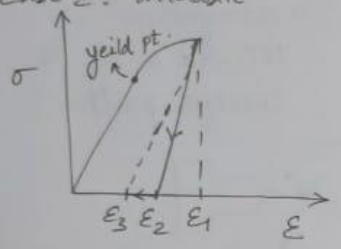
case I: "anelastic"

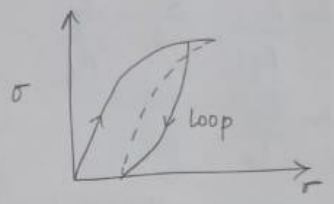


$\varepsilon_1 - \varepsilon_2 \rightarrow$ elastic recovery

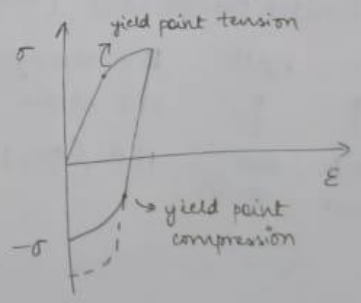
$\varepsilon_2 - \varepsilon_2 \rightarrow$ anelastic recovery

(caused due to temp. or material prop.)

case II: "hysteresis"



case III: "Bauschinger effect"



$$\begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix} = \begin{bmatrix} \bar{\sigma} & 0 & 0 \\ 0 & \bar{\sigma} & 0 \\ 0 & 0 & \bar{\sigma} \end{bmatrix} + \underbrace{\begin{bmatrix} \sigma_1 - \bar{\sigma} & 0 & 0 \\ 0 & \sigma_2 - \bar{\sigma} & 0 \\ 0 & 0 & \sigma_3 - \bar{\sigma} \end{bmatrix}}_{\text{Daiabatric stress}}$$

$$I_1 = (\sigma_1 - \bar{\sigma}) + (\sigma_2 - \bar{\sigma}) + (\sigma_3 - \bar{\sigma})$$

$$I_1 = 0 \rightarrow \boxed{\text{pure shear}}$$

$$U = \frac{1}{2}\sigma\varepsilon = \text{energy per unit volume}$$

Total strain energy $= U_{total} = \frac{1}{2}\sigma_1\varepsilon_1 + \frac{1}{2}\sigma_2\varepsilon_2 + \frac{1}{2}\sigma_3\varepsilon_3$

$$U_{total} = \frac{1}{2}\sigma_1 \frac{1}{E}\left(\sigma_1 - \nu(\sigma_2 + \sigma_3)\right)$$
$$+ \frac{1}{2}\sigma_2 \frac{1}{E}\left(\sigma_2 - \nu(\sigma_1 + \sigma_3)\right)$$
$$+ \frac{1}{2}\sigma_3 \frac{1}{E}\left(\sigma_3 - \nu(\sigma_2 + \sigma_1)\right)$$
$$= \frac{1}{2E}\left(\sigma_1^2 + \sigma_2^2 + \sigma_3^2\right) - \frac{\nu}{E}\left(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_1\sigma_3\right)$$

$$U_{hydrostatic} = \frac{1}{2E}\left(3\bar{\sigma}^2\right) - \frac{\nu}{E}\left(3\bar{\sigma}^2\right)$$

$$= \frac{3(1-2\nu)}{2E}\bar{\sigma}^2$$

$$U_{diabatric} = U_{total} - U_{hydrostatic}$$

$$= \frac{1}{2E}\left(\sigma_1^2+\sigma_2^2+\sigma_3^2\right) - \frac{\nu}{E}\left(\sigma_1\sigma_2+\sigma_2\sigma_3+\sigma_1\sigma_3\right)$$

$$- \frac{3(1-2\nu)}{2E}\left(\frac{1}{3}\left(\sigma_1+\sigma_2+\sigma_3\right)\right)^2$$

$$= \frac{1}{2E}\left(\sigma_1^2+\sigma_2^2+\sigma_3^2\right) - \frac{\nu}{E}\left(\sigma_1\sigma_2+\sigma_2\sigma_3+\sigma_1\sigma_3\right)$$

$$- \frac{3(1-2\nu)}{2E}\left(\frac{1}{9}\left(\left(\sigma_1^2+\sigma_2^2+\sigma_3^2\right)+2(\quad)\right)\right)$$

$$= \frac{\left(\sigma_1^2+\sigma_2^2+\sigma_3^2\right)}{6G}\left(\frac{9}{2E} - \frac{3(1-2\nu)}{2E}\left(\frac{1}{9}\right)\right)$$

$$- \frac{\left(\sigma_1\sigma_2+\sigma_2\sigma_3+\sigma_{13}\right)}{6G}\left(\frac{9\nu}{E} + \frac{3(1-2\nu)}{2E}\right)$$

$$= \frac{1}{6G}\left(\left(\sigma_1^2+\sigma_2^2+\sigma_3^2\right) - \left(\sigma_1\sigma_2+\sigma_2\sigma_3+\sigma_1\sigma_3\right)\right)$$

Yielding

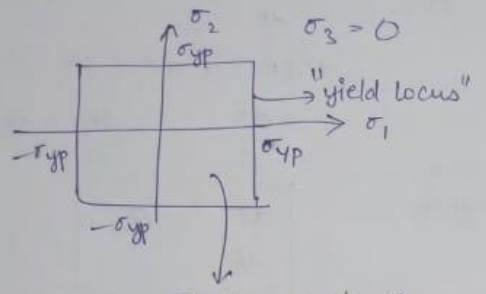
$\sigma_1, \sigma_2, \sigma_3 \Rightarrow$ principal stress

* $\boxed{\text{Rankin's theory}} \Rightarrow$ good for ~~ductile~~ brittle material

assume $|\sigma_1| > |\sigma_2| > |\sigma_3|$

→ also called "Max principal stress theory"

when, $|\sigma_1| > \sigma_{yp}$, material starts deforming.



Inside: no plastic deformation (no yielding)

Hydrostatic stress $\Rightarrow$ responsible for volume change

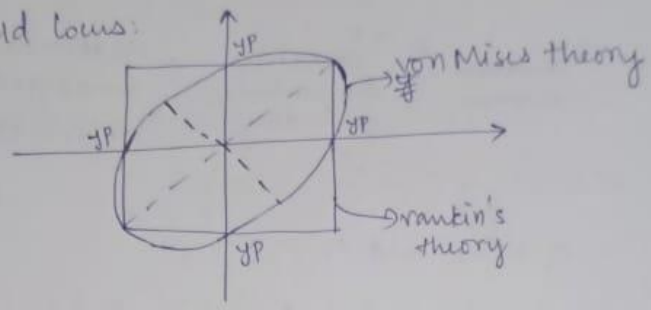
$$\hookrightarrow \begin{bmatrix} \bar{\sigma} & 0 & 0 \\ 0 & \bar{\sigma} & 0 \\ 0 & 0 & \bar{\sigma} \end{bmatrix} \quad \text{where } \bar{\sigma} = \frac{1}{3}(\sigma_1 + \sigma_2 + \sigma_3)$$

To get yield locus,

→ If $\sigma_3 = 0$,

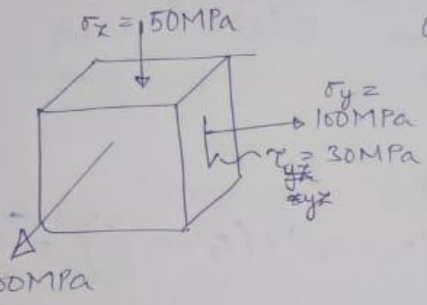
$$\sigma_1^2 - \sigma_1\sigma_2 + \sigma_2^2 = \sigma_{yp}^2 \Big\} \text{Ellipse}$$

Yield locus:



Q.

$\sigma_{yp} = 410\,MPa$



$$\sigma_{ij} = \begin{bmatrix} 200 & 0 & 0 \\ 0 & 100 & 30 \\ 0 & 30 & -50 \end{bmatrix}$$

$I_1 = 200 + 100 - 50 = 250$

$I_2 = \cancel{20000 - 10000 - 5000}$

$= 5000$

$I_3 =$

$$U_{\substack{\text{deviatoric}\\ \text{distorsion}}} = \frac{1}{12G}\left((\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_1-\sigma_3)^2\right)$$

When $U_{\substack{\text{deviatoric}\\ \text{distortion}}} = U_{\substack{\text{deviatoric}\\ \text{for uniaxial}\\ \text{loading}}}$, material undergoes deformation.

$$\sigma_1 = \sigma_{yp} \text{ and } \sigma_2 = \sigma_3 = 0$$

$$U_{\substack{\text{deviatoric}\\ \text{for uniaxial}\\ \text{loading}}} = \frac{1}{12G}\left(2\sigma_{yp}^2\right) = \frac{\sigma_{yp}^2}{6G}$$

von Mises theory:

$$\frac{1}{12G}\left((\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2\right) = \frac{\sigma_{yp}^2}{6G}$$

$$\sigma_{yp} = \frac{1}{\sqrt{2}}\left((\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2\right)^{1/2}$$

von Mises criteria / shear strain energy criteria
(for ductile materials)